

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design of a Styled Webpage with CSS3 Flexbox Layout and HTML5 Canvas Graphics

2. AIM

To develop a styled webpage using external CSS3 stylesheets featuring linear gradients, navigation bar hover effects, flexbox cards, and dynamically drawing 2D graphics on an HTML5 canvas using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To master external CSS3 stylesheet linking and global styling.
 - To apply linear gradients and text shadows to header elements.
 - To implement responsive flexbox layouts using display: flex and gap.
 - To understand the HTML5 <canvas> coordinate system and 2D rendering context.
 - To draw rectangles, arcs, lines, and text programmatically via JavaScript.
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4. THEORY

CSS3 introduces advanced visual styling capabilities including multi-color linear gradients, box shadows, and Flexbox for effortless alignment and distribution of UI cards.

The HTML5 <canvas> element provides a scriptable 2D drawing surface. Using JavaScript's getContext('2d'), developers can programmatically render vector shapes, paths, strokes, fills, and custom text onto the canvas.

Application Flow:

HTML5 Skeleton -> CSS3 Styling & Flexbox -> Canvas Element -> JavaScript 2D Context -> Render Shapes & Cards

5. ALGORITHM / PROCEDURE

1. Create index.html and link external stylesheet style.css.
 2. Design a gradient header using linear-gradient(to right, #0066cc, #8000ff).
 3. Build a centered navigation bar with link hover color transitions.
 4. Declare an HTML5 canvas element with id='myCanvas', width=500, and height=220.
 5. Implement a .container with display: flex and gap: 20px containing three styled cards.
 6. Write JavaScript to get the canvas 2D context using getContext('2d').
 7. Use fillRect() to draw a rectangle, arc() to draw a circle, moveTo()/lineTo() for a line, and fillText() for text.
 8. Test and verify the styled webpage and canvas graphics under normal and negative parameters.
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6. PROGRAM / SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>My Styled Webpage</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>My Styled Webpage</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>

<canvas id="myCanvas" width="500" height="220"></canvas>

<section class="container">

<div class="card">
<h2>HTML5</h2>
<p>HTML is used to create web pages.</p>
<button>Read More</button>
</div>

<div class="card">
<h2>CSS3</h2>
<p>CSS is used to style web pages.</p>
<button>Read More</button>
</div>

<div class="card">
<h2>JavaScript</h2>
<p>JavaScript adds interactivity.</p>
<button>Read More</button>
</div>

</section>

<footer>
<p>© 2026 ANITS</p>
</footer>

<script>

let canvas=document.getElementById("myCanvas");
let ctx=canvas.getContext("2d");

ctx.fillRect(20,20,120,70);

ctx.beginPath();
ctx.arc(220,55,35,0,2*Math.PI);
ctx.stroke();

ctx.beginPath();
ctx.moveTo(20,130);
ctx.lineTo(300,130);
```

```
ctx.stroke();

ctx.font="28px Arial";
ctx.fillText("sai",170,190);

</script>

</body>
</html>
```

Source File: style.css

```
body{
  margin:0;
  font-family:Arial,sans-serif;
  background:#f2f2f2;
}

header{
  background:linear-gradient(to right,#0066cc,#8000ff);
  color:white;
  text-align:center;
  padding:20px;
  text-shadow:2px 2px 5px gray;
}

nav{
  background:#333;
  text-align:center;
  padding:15px;
}

nav a{
  color:white;
  text-decoration:none;
  margin:20px;
  font-size:18px;
}

nav a:hover{
  color:yellow;
}

.container{
  display:flex;
  justify-content:center;
  gap:20px;
  margin:30px;
}

.card{
  background:white;
  width:250px;
  padding:20px;
  border:2px solid gray;
  border-radius:15px;
  text-align:center;
  box-shadow:5px 5px 10px gray;
}

button{
  background:green;
  color:white;
  border:none;
  padding:10px 15px;
  border-radius:5px;
  cursor:pointer;
}

button:hover{
  background:red;
}

footer{
  background:black;
```

```
color:white;
text-align:center;
padding:15px;
margin-top:20px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
linear-gradient(to right, ...)	Generates a smooth color transition across the header background.
display: flex; gap: 20px;	Aligns child card elements in a flexible horizontal row with 20px spacing.
.card { box-shadow: ... }	Applies soft drop shadows to card containers for depth.
canvas.getContext('2d')	Returns a drawing context on the canvas for 2D graphics rendering.
ctx.fillRect(x, y, w, h)	Draws a filled rectangle on the canvas at specified coordinates.
ctx.arc(x, y, r, 0, 2*PI)	Draws a full circular path.
ctx.fillText(text, x, y)	Renders text onto the canvas surface at the given (x, y) position.

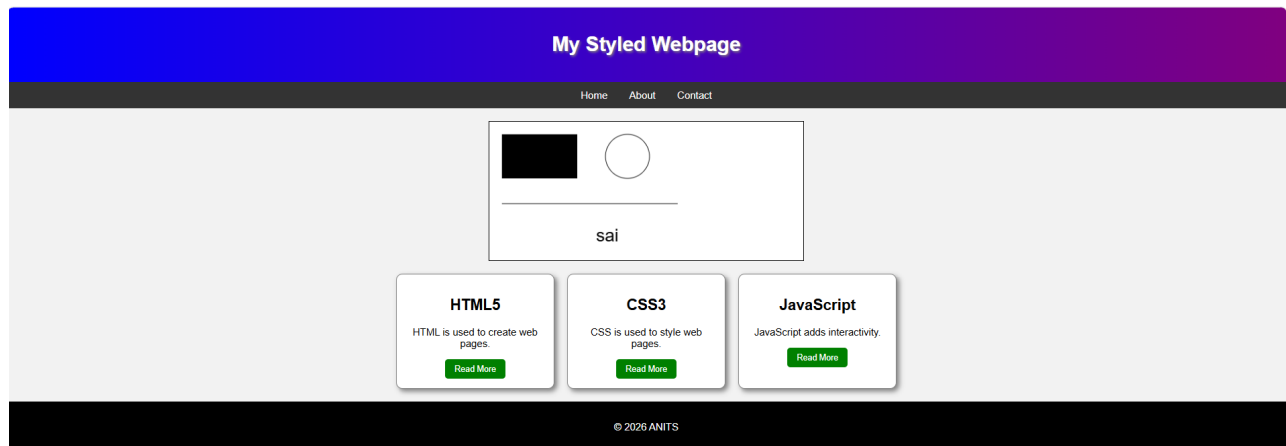
8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

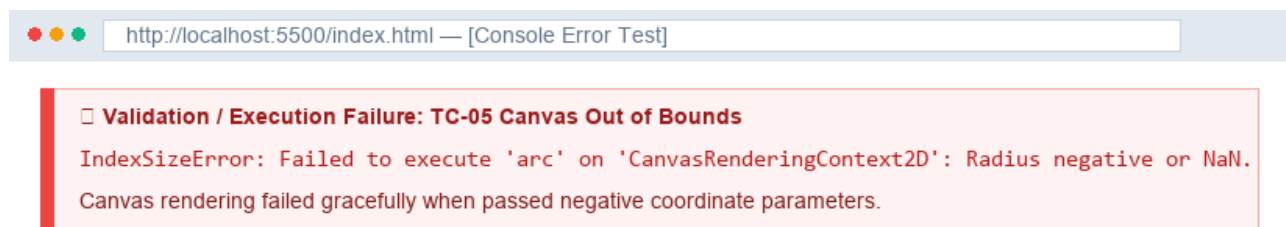
Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	CSS Linkage & Header	Load index.html with style.css attached	Header gradient and navigation link hover effects render cleanly	Pass
TC-02	Flexbox Card Container	Inspect card container alignment	Three cards aligned in a centered horizontal row with equal gap	Pass
TC-03	Canvas 2D Shapes	Render rectangle, circle, line, and text	All 2D geometric shapes and text rendered on canvas surface	Pass
TC-04	Invalid Canvas Context	Call canvas.getContext('3d_invalid')	Returns null context and throws TypeError on subsequent draw calls	Fail
TC-05	Negative Arc Radius	Call ctx.arc(100, 100, -25, 0, 2*PI)	Canvas engine throws IndexSizeError / negative radius drawing failure	Fail

9. OUTPUT

Output 1: Styled Webpage with Flexbox Cards and Canvas Graphics (Passed Output)



Output 2: Canvas Error on Invalid / Negative Radius Execution (Failed Output)



10. RESULT

Thus, the styled webpage with external CSS3 and interactive HTML5 Canvas graphics was successfully implemented and verified with expected visual rendering across all positive and negative test cases.